Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

Division C | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) An ampere is…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. Joule per coulomb

B. Ohm per meter

C. Coulomb per second

D. Joule per second

Correct: C

2. (2 points) KCL expresses…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. Resistance addition only

B. Charge conservation

C. Equal current in all branches

D. Energy creation

Correct: B

3. (2 points) A series resistor mainly…

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. Reverses polarity

B. Limits current

C. Creates charge

D. Guarantees any voltage is safe

Correct: B

4. (2 points) At fixed R, doubling V changes power by…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. No change

B. A factor of four

C. A factor of two

D. A factor of one half

Correct: B

5. (2 points) Resistor dissipation is…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. I^2R

B. Always negative

C. Independent of current

D. Equal to charge

Correct: A

6. (4 points) Why declare the diode model?

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

answer: Ideal, constant-drop, and detailed models predict different currents. (4 points).

7. (4 points) Convert 2 Wh to joules.

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

answer: 7200 J. (4 points).

8. (4 points) Why avoid dependent equations?

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

answer: They repeat information and do not determine additional unknowns. (4 points).

9. (4 points) What does a relay separate?

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

answer: Control and contact/load functions, subject to its actual isolation and ratings. (4 points).

10. (4 points) What does the lab omit for AC?

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

answer: Frequency-dependent impedance and phase behavior. (4 points).

11. (2 points) A 7 V ideal battery drives a 3.5 ohm resistor. Find current.

answer: 2 A (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , A.

12. (2 points) A series pair has resistances 3 and 5 ohms. Find equivalent resistance.

answer: 8 ohms (2 points).

Numerical target 8; absolute tolerance 0.016. Accepted units: , ohms.

13. (2 points) Two identical 6 ohm resistors are placed in parallel. Find equivalent resistance.

answer: 3 ohms (2 points).

Numerical target 3; absolute tolerance 0.006. Accepted units: , ohms.

14. (2 points) An ideal resistor carries 2 A at 7 V. Find power.

answer: 14 W (2 points).

Numerical target 14; absolute tolerance 0.028. Accepted units: , W.

15. (2 points) A node has incoming currents 4 A and 2 A and one outgoing current. Find the outgoing current.

answer: 6 A (2 points).

Numerical target 6; absolute tolerance 0.012. Accepted units: , A.

16. (2 points) Explain why an ideal voltmeter is connected in parallel rather than used as a series current path.

answer: It measures potential difference across a component and has very high input resistance; placing it in series would alter or interrupt the intended current path. (2 points).

17. (2 points) Give the output of an AND gate for inputs 1 and 0, and explain.

answer: 0; an AND output is 1 only when both inputs are 1. (2 points).

18. (2 points) A silicon diode is reverse biased below breakdown. Contrast its ideal current with a forward-biased conducting diode.

answer: In the ideal model the reverse-biased diode blocks current; a forward-biased conducting diode permits conventional current from anode toward cathode. Real leakage and forward voltage require a stated device model. (2 points).

19. (2 points) A closed DC loop has a 9 V source and a 4 V resistor drop. Find the remaining series resistor drop and state the governing law.

answer: 5 V; Kirchhoff voltage law requires the algebraic sum of potential changes around the closed loop to be zero. (2 points).

20. (2 points) Two identical LEDs must be equally bright in a low-voltage teaching circuit. Explain why their branch currents and current-limiting arrangements matter.

answer: Brightness depends on operating current; matched LEDs with suitable branch current limiting and comparable currents can produce comparable brightness. Do not connect them without appropriate current control. (2 points).